



Group 9 - Wine Tasters

Tiffany Peng, Marvin Perez De Pablo, Cameron Fong,
Nancy Lopez, Axel Galicia Lona



Introduction

Question of Interest:

Can we accurately predict the density of wine in our dataset?

- In order to answer our question, we will be conducting a linear regression analysis
- We expect to find relationships and determine how much our predictor variables can explain the variation in our response variable.

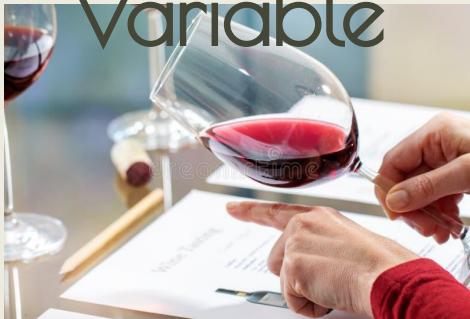
Data:

- The dataset is from Kaggle
- Original dataset has 1599 observations and 12 variables
 - No N/A's or outliers





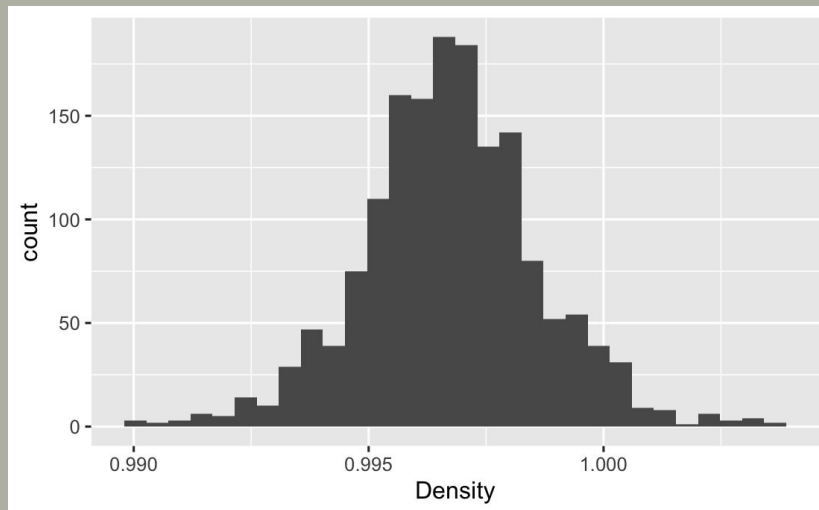
Response Variable



- Our response variable (y) is density, which is measured in grams per milliliter (g/mL).
- Density is a measure of how much mass is contained in a given volume of a substance.



EDA of Density



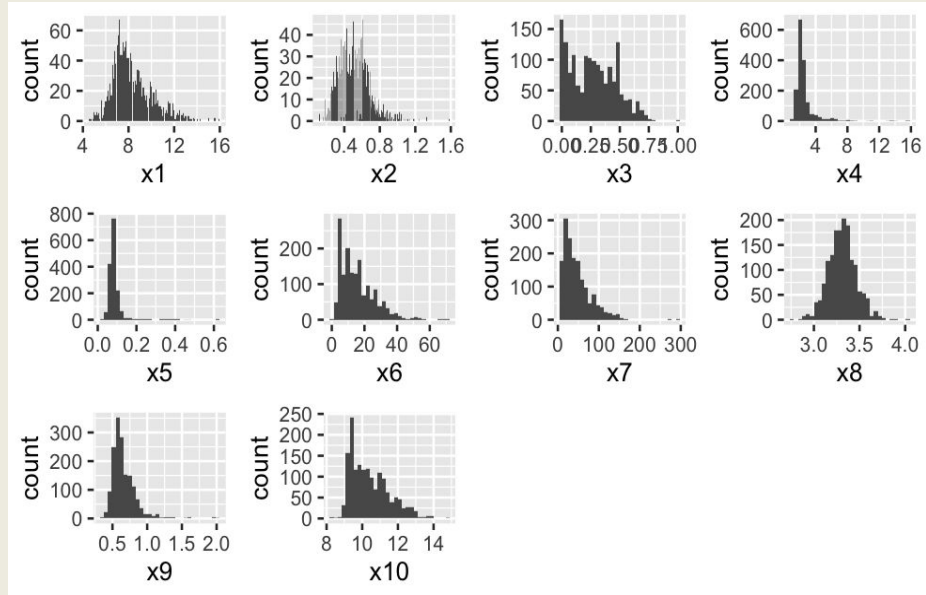
- Density seems to follow an approximately normal distribution.
- This means no transformations needed.

Predictor

Variables

- X1 = fixed acidity (grams of acid per liter)
- X2 = volatile acidity (grams of acid per liter)
- X3 = citric acid (mg/L)
- X4 = residual sugar (g/L)
- X5 = chlorides (g/L)
- X6 = free sulfur dioxide (mL)
- X7 = total sulfur dioxide (mL)
- X8 = pH (concentration of hydrogen ions per liter)
- X9 = sulphates
- X10 = alcohol (grams)

EDA of Predictors



Our predictor variables display a bell-shaped distribution with a slight right skewness, but they are approaching normality. Hence, transformations will only be considered if necessary later in the process.



Initial Results

Our initial model included all predictors,

$$\hat{Y} = 9.7e-01 + 9.421e-04*x1 + 7.189e-04*x2 + 5.849e-05*x3 + 4.128e-04*x4 + 1.818e-03*x5 - 8.575e-06*x6 + 2.654e-06*x7 + 5.068e-03*x8 + 1.311e-03*x9 - 9.253e-04*x10$$

It also showed that there is no potential multicollinearity since all VIF values are under 5.

	x1	x2	x3	x4	x5	x6	x7	x8	x9
2.975491	1.759879	3.127791	1.099433	1.468893	1.948691	2.173240	2.239412	1.341524	
x10									
1.299603									

This model is pretty good, but we want to explore different regression methods to find a better model with a lower MSE and higher adjusted R^2 .



```
Call:
lm(formula = density ~ ., data = winequality)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.830e-03 -4.256e-04 -2.078e-05  4.240e-04  2.884e-03
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.793e-01  7.107e-04 1378.045 < 2e-16 ***
fixed.acidity  9.421e-04  1.863e-05  50.571 < 2e-16 ***
volatile.acidity 7.189e-04  1.393e-04  5.160 2.78e-07 ***
citric.acid    5.849e-05  1.707e-04  0.343 0.731944
residual.sugar 4.128e-04  1.398e-05 29.516 < 2e-16 ***
chlorides     1.818e-03  4.842e-04  3.755 0.000180 ***
free.sulfur.dioxide -8.575e-06  2.509e-06 -3.417 0.000649 ***
total.sulfur.dioxide 2.654e-06  8.427e-07  3.149 0.001667 **
pH            5.068e-03  1.823e-04 27.806 < 2e-16 ***
sulphates     1.311e-03  1.285e-04 10.201 < 2e-16 ***
alcohol      -9.253e-04  2.012e-05 -45.998 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.0007517 on 1588 degrees of freedom
Multiple R-squared:  0.8424,    Adjusted R-squared:  0.8414
F-statistic: 848.6 on 10 and 1588 DF, p-value: < 2.2e-16
```

Model Building

- Used `MASS` library to look into the interactions of predictors
- The model showed good results
 - However it was too complex
 - Had over 30 interactions
 - Would not be useful in real-life practice
- With this, we will be proceeding with different regression methods
 - Stepwise Regression
 - Polynomial
 - More simple and realistic model
 - Higher adjusted R^2 and lower MSE

```
```{r results='hide'}
#> using the mass library to get a model with interactions
full_model <- lm(density ~ (.)^2, data = winequality)
step_model_interactions <- stepAIC(full_model, direction = "both")
```{r}
# Training and Test MSE for interactions model, R^2: 0.8804
interactions.model <- lm(density ~ fixed.acidity + volatile.acidity + citric.acid +
  residual.sugar + chlorides + free.sulfur.dioxide + total.sulfur.dioxide +
  pH + sulphates + alcohol + fixed.acidity:volatile.acidity +
  fixed.acidity:chlorides + fixed.acidity:free.sulfur.dioxide +
  fixed.acidity:total.sulfur.dioxide + fixed.acidity:pH + fixed.acidity:sulphates +
  fixed.acidity:alcohol + volatile.acidity:residual.sugar +
  volatile.acidity:chlorides + volatile.acidity:free.sulfur.dioxide +
  volatile.acidity:total.sulfur.dioxide + volatile.acidity:sulphates +
  volatile.acidity:alcohol + citric.acid:residual.sugar + citric.acid:chlorides +
  citric.acid:pH + citric.acid:sulphates + citric.acid:alcohol +
  residual.sugar:total.sulfur.dioxide + residual.sugar:pH +
  residual.sugar:sulphates + chlorides:total.sulfur.dioxide +
  chlorides:pH + chlorides:sulphates + chlorides:alcohol +
  free.sulfur.dioxide:total.sulfur.dioxide + free.sulfur.dioxide:pH +
  free.sulfur.dioxide:sulphates + free.sulfur.dioxide:alcohol +
  total.sulfur.dioxide:pH + total.sulfur.dioxide:alcohol +
  pH:alcohol, train)
mean((train$density - predict(interactions.model))^2)
mean((test$density - predict(interactions.model, newdata = test))^2)
```
```

[1] 4.123041e-07  
[1] 4.56874e-07

# Comparing Models

Training: 70%    Testing: 30%

## 1. Full Model (All predictors)

- Train MSE:  $5.53e-07$
- Test MSE:  $5.84e-07$
- Adj.  $R^2$ : 0.8403

## 2. Stepwise Full Model

- Train MSE:  $5.54e-07$
- Test MSE:  $5.82e-07$
- Adj.  $R^2$ : 0.8415

## 3. Stepwise Interactions Model

- Train MSE:  $4.18e-07$
- Test MSE:  $4.64e-07$
- Adj.  $R^2$ : 0.8789

## 4. Polynomial Model (fixed acidity<sup>2</sup>, alcohol<sup>2</sup>)

- Train MSE:  $5.46e-07$
- Test MSE:  $5.69e-07$
- Adj.  $R^2$ : 0.8422

## 5. Interactions Model (All possible two predictor interactions)

- Train MSE:  $4.12e-07$
- Test MSE:  $4.57e-07$
- Adj.  $R^2$ : 0.8804



# Multicollinearity Analysis

## 4. Polynomial Model (fixed.acidity<sup>2</sup>, alcohol<sup>2</sup>)

- Train MSE: 5.46e-07
- Test MSE: 5.69e-07
- Adj. R<sup>2</sup>: 0.8422

|               |                  |             |                                |                          |                     |                      |
|---------------|------------------|-------------|--------------------------------|--------------------------|---------------------|----------------------|
| fixed.acidity | volatile.acidity | citric.acid | residual.sugar                 | chlorides                | free.sulfur.dioxide | total.sulfur.dioxide |
| 71.693291     | 1.754917         | 3.069993    | 1.090809                       | 1.551016                 | 2.137826            | 2.396904             |
| pH            | sulphates        | alcohol     | I(fixed.acidity <sup>2</sup> ) | I(alcohol <sup>2</sup> ) |                     |                      |
| 2.345508      | 1.391201         | 322.588700  | 65.071823                      | 321.405739               |                     |                      |

## Model we chose: Polynomial Model

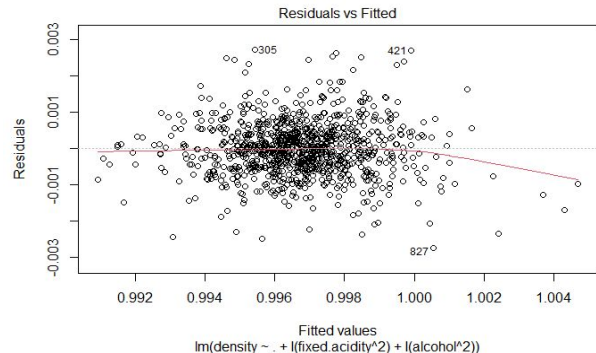
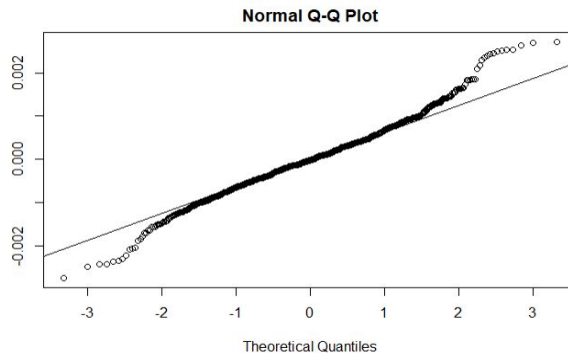
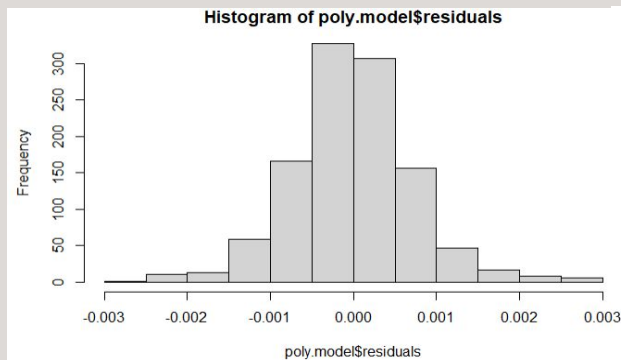
- Vif → removed alcohol, fixed.acidity
- Caused minor dip in Adj. R<sup>2</sup>: 0.8253
- Higher MSE than before, train: 6.05e-07, test: 6.66e-07

|                  |                                |                          |           |                     |                      |          |
|------------------|--------------------------------|--------------------------|-----------|---------------------|----------------------|----------|
| volatile.acidity | citric.acid                    | residual.sugar           | chlorides | free.sulfur.dioxide | total.sulfur.dioxide | pH       |
| 1.738966         | 2.987644                       | 1.090474                 | 1.525907  | 2.097986            | 2.296876             | 2.135277 |
| sulphates        | I(fixed.acidity <sup>2</sup> ) | I(alcohol <sup>2</sup> ) |           |                     |                      |          |
| 1.385746         | 2.731279                       | 1.313439                 |           |                     |                      |          |



# Model Assumptions

Model =  $\text{lm}(\text{density} \sim \text{fixed acidity}^2 + \text{alcohol}^2 + .)$



## Durbin-watson test

```
data: poly.model
DW = 1.9805, p-value = 0.3716
alternative hypothesis: true autocorrelation is greater than 0
```

## studentized Breusch-Pagan test

```
data: poly.model
BP = 121.96, df = 12, p-value < 2.2e-16
```

- Residuals histogram looks normal thus variance is normally distributed
- QQ plot shows normality of data
- Residual vs fitted plot look random → Constant variance and linear model appropriate
- Durbin Watson test is approximately 2, no autocorrelation in the data
- Breusch pagan test p-value less than 0.05 → Heteroscedasticity is present (the residuals are not distributed with equal variance)

# Final Model & Interpretations

$$\hat{Y} = 1.176e^{-03}x_1 + 6.227e^{-04}x_2 - 1.532e^{-04}x_3 + 4.224e^{-04}x_4 + 2.123e^{-03}x_5 - 1.010e^{-05}x_6 + 3.075e^{-06}x_7 + 5.066e^{-03}x_8 + 1.215e^{-03}x_9 + 2.234e^{-04}x_{10} - 1.200e^{-05}x_{11} - 5.151e^{-05}x_{12} + 9.721e^{-01}$$

Where:  $\hat{Y}$ : Density,  $x_1$ : fixed.acidity,  $x_2$ : volatile.acidity,  $x_3$ : citric.acid,  $x_4$ : residual.sugar,  $x_5$ : chlorides,  $x_6$ : free.sulfur.dioxide,  $x_7$ : total.sulfur.dioxide,  $x_8$ : pH,  $x_9$ : sulphates,  $x_{10}$ : alcohol,  $x_{11}$ :  $I(\text{fixed.acidity}^2)$ ,  $x_{12}$ :  $I(\text{alcohol}^2)$

## Interpretations (assuming all else is held constant):

- For every one unit increase in fixed.acidity, density increases by  $1.176e^{-03}$
- For every one unit increase in volatile.acidity, density increases by  $6.227e^{-04}$
- For every one unit increase in citric.acid, density decreases by  $1.532e^{-04}$
- For every one unit increase in residual.sugar, density increases by  $4.224e^{-04}$
- For every one unit increase in chlorides, density increases by  $2.123e^{-03}$
- For every one unit increase in free.sulfur.dioxide, density decreases by  $1.010e^{-05}$
- For every one unit increase in total.sulfur.dioxide, density increases by  $3.075e^{-06}$
- For every one unit increase in pH, density increases by  $5.066e^{-03}$
- For every one unit increase in sulphates, density increases by  $1.215e^{-03}$
- For every one unit increase in alcohol, density increases by  $2.234e^{-04}$
- For every one unit increase in  $I(\text{fixed.acidity}^2)$ , density decreases by  $1.200e^{-05}$
- For every one unit increase in  $I(\text{alcohol}^2)$ , density decreases by  $5.151e^{-05}$

# Conclusion & Limitations

- The model with interactions may have been a good model, but it was too complex with over 30 interactions and was not easily interpretable
- Current polynomial model has good predictability accuracy, good for inference
- Citric acid has the largest negative relationship with density at  $1.532\text{e-}04$  density per unit of citric acid
- Total sulfur dioxide has the largest positive relationship with density at  $3.075\text{e-}06$  density per unit of sulfur dioxide
- Multicollinearity issues (could use other regression methods of analysis)
- Heteroskedasticity could cause issues with standard errors of estimates



# Thank you!

## Questions?



## Contributions:

- Tiffany Peng: Created slides and helped with project objectives
- Nancy Lopez: Helped created EDA and interpretations
- Axel Galicia Lona - Create model and checked for multicollinearity
- Cameron Fong - Helped create model and interpretations
- Marvin Perez de Pablo - Helped created EDA and interpretations